

Understanding Bitcoin mining

Mining is the verification of bitcoin transactions. As shown in Figure 1, let us suppose Hema wants to buy a computer from David, using Bitcoin. To make sure her Bitcoin is genuine, miners begin to verify the transaction by solving a proof-of-work problem, which is an incredibly complex math problem. The transaction is approved only after miners look into the transaction history of Hema to verify if she has enough valid Bitcoins to process the transaction. Once approved, this transaction is updated in the public ledger as a new block and the message is propagated to all the users in that blockchain network. The miner who validates the transaction first gets to place the next block into the blockchain. To incentivise mining, the miners are rewarded with newly released Bitcoins. This provides a smart way to issue the currency and also creates an incentive for more people to mine.

Blockchain technology – how it works

Blockchain technology is based on a decentralised, distributed ledger system. Working with blockchain can be understood based on how a new block is added to the blockchain network. The steps that are carried out during the operation in the blockchain network are depicted in Figure 2 and described below.

Step 1: Let us suppose User A wants to send 500 Bitcoins to User B. Then, A requests a transaction.

Step 2: The User A's node starts a transaction by first creating and then digitally signing it with A's private key (created via cryptography). A

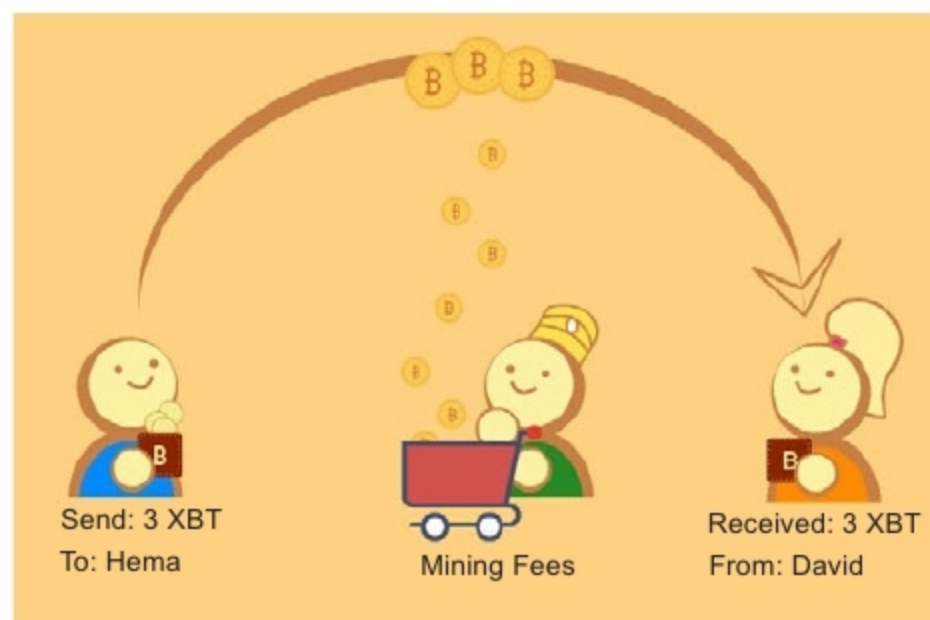


Figure 1: Bitcoin mining in a blockchain

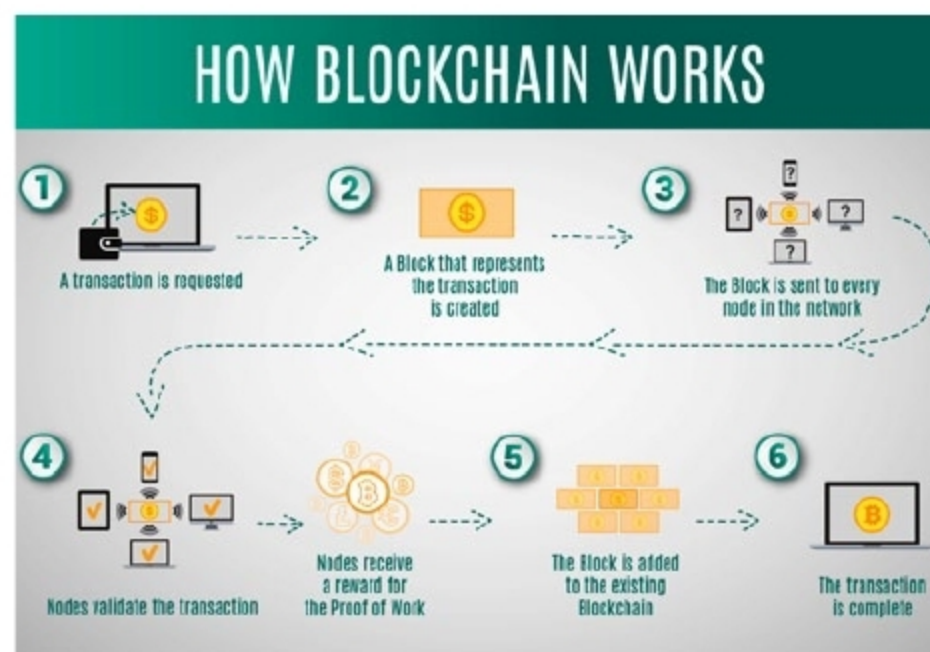


Figure 2: How blockchain technology works

transaction can represent various actions in a blockchain.

Step 3: The transaction is propagated (flooded) by using a flooding protocol, called a gossip protocol, to peers who validate the transaction based on pre-set criteria. Usually, more than one node, called miners, can verify the transaction.

Step 4: Once the transaction is validated, it is included in a block, which is then propagated onto the network. At this point, the transaction is considered confirmed.

Step 5: The newly-created block now becomes part of the ledger and the next block links itself cryptographically back to this block. This link is a hash pointer.

Step 6: When that new block is added to the blockchain, it becomes publicly available for anyone to view. The blockchain transaction is now complete and the ledger is updated.

Blockchain and security

One of the important aspects of blockchain technology is that no one can manipulate the recorded transaction data. This is because, in the blockchain network, every transaction is propagated between nodes using a decentralised distributed ledger in a peer-to-peer (P2P) manner. In addition, an advanced level of encryption, along with the hashing mechanism (SHA-256), is used. Thus, this technology is highly secure as it is almost impossible to tamper with or hack the transactions.

The importance of open source software for blockchain technology

Open source software is collaboratively produced, shared freely, published transparently and developed for the good of the community. It is not the property of a single company or person. It ensures better transparency and offers new possibilities. Open source software provides several key benefits for blockchain projects, as listed below:

- Enables developers to build purpose-specific Bitcoin-compatible applications (e.g., a wallet app for smartphones).
- Enables the building of a new cryptocurrency that ceases to